

# Magnetostatics

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## Problem 5.1

## Problem 5.2

## Problem 5.3

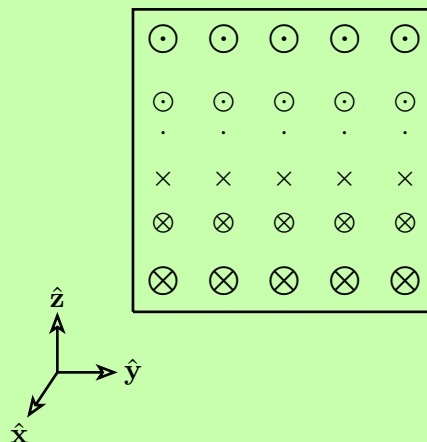
## Problem 5.4

Suppose that the magnetic field in some region has the form

$$\mathbf{B} = kz\hat{\mathbf{x}}$$

(where  $k$  is a constant). Find the force on a square loop (side  $a$ ), lying in the  $yz$  plane and centered at the origin, if it carries a current  $I$ , flowing counterclockwise, when you look down the  $x$  axis.

### Solution:



The force on a wire carrying a current  $\mathbf{I}$  is

$$\mathbf{F} = \int I(\mathrm{d}\mathbf{l} \times \mathbf{B})$$

With current  $I$ , the forces on each of the four lines are, starting with the top line:

$$z = \frac{a}{2}, \quad y : \frac{a}{2} \rightarrow -\frac{a}{2}$$

$$B = k\frac{a}{2}\hat{\mathbf{x}}, \quad \mathrm{d}\mathbf{l} = \hat{\mathbf{y}} \mathrm{d}y$$

$$\mathbf{F} = Ik\frac{a}{2} \int_{\frac{a}{2}}^{-\frac{a}{2}} (\hat{\mathbf{y}} \times \hat{\mathbf{x}}) \mathrm{d}y = -Ik\frac{a}{2}\hat{\mathbf{z}} \int_{\frac{a}{2}}^{-\frac{a}{2}} \mathrm{d}y = Ik\frac{a}{2}\hat{\mathbf{z}} \int_{-\frac{a}{2}}^{\frac{a}{2}} \mathrm{d}y$$

↓

$$\mathbf{F} = \frac{Ika^2}{2}\hat{\mathbf{z}}$$

Bottom line:

$$B = -k\frac{a}{2}\hat{\mathbf{x}}, \quad \mathrm{d}\mathbf{l} = \hat{\mathbf{y}} \mathrm{d}y, y : -\frac{a}{2} \rightarrow \frac{a}{2}$$

$$\mathbf{F} = -Ik\frac{a}{2} \int_{-\frac{a}{2}}^{\frac{a}{2}} (\hat{\mathbf{y}} \times \hat{\mathbf{x}}) \mathrm{d}y = Ik\frac{a}{2}\hat{\mathbf{z}} \int_{-\frac{a}{2}}^{\frac{a}{2}} \mathrm{d}y$$

↓

$$\mathbf{F} = \frac{Ika^2}{2}\hat{\mathbf{z}}$$

The forces on the left and right wires are zero due to symmetry (there is equal parts of these wires in a  $+\hat{\mathbf{x}}$  magnetic field as  $-\hat{\mathbf{x}}$  magnetic field).

Total is top + Bottom

$$\mathbf{F}_{\text{tot}} = \mathbf{F}_{\text{top}} + \mathbf{F}_{\text{bot}} = \frac{Ika^2}{2}\hat{\mathbf{z}} + \frac{Ika^2}{2}\hat{\mathbf{z}} = \boxed{Ika^2\hat{\mathbf{z}}}$$

□

## Problem 5.5

A current  $I$  flows down a wire of radius  $a$ .

- (a) If it is uniformly distributed over the surface, what is the surface current density  $K$  ?
- (b) If it is distributed in such a way that the volume current density is inversely proportional to the distance from the axis, what is  $J(s)$ ?

**Solution:**

(a) A wire of radius  $a$  has circumference  $2\pi a$  so the current passing through each infinitesimal part of the surface of the wire (circumference) is simply

$$K = \frac{I}{2\pi a}$$

(b) We want the volume current density to be

$$J = \frac{k}{s}$$

where  $k$  is the proportionality constant. If we were to integrate  $J(s)$  over a cross section of the wire, we should recover  $I$

$$I = \int J(s) = \int_0^a \int_0^{2\pi} \frac{k}{s} s \, ds \, d\phi = \int_0^a \int_0^{2\pi} k \, ds \, d\phi \rightarrow I = ka2\pi$$

$$\downarrow$$

$$k = \frac{I}{2\pi a}$$

so

$$J(s) = \frac{I}{2\pi as}$$



□

## Problem 5.6

## Problem 5.7

## Problem 5.8

- (a) Find the magnetic field at the center of a square loop, which carries a steady current  $I$ . Let  $R$  be the distance from center to side.
- (b) Find the field at the center of a regular  $n$ -sided polygon, carrying a steady current  $I$ . Again, let  $R$  be the distance from the center to any side.
- (c) Check that your formula reduces to the field at the center of a circular loop, in the limit  $n \rightarrow \infty$ .

### Solution:

(a) We can find the field produced by one of the sides, and due to symmetry the result is four times of that. Calculating the right wire gives

$$\begin{aligned} x &= R, \quad y : -R \rightarrow R, \quad d\mathbf{l}' = \hat{\mathbf{y}} dy \\ \mathbf{r} &= -R\hat{\mathbf{x}} - y\hat{\mathbf{y}}, \quad r = \sqrt{R^2 + y^2} \\ \hat{\mathbf{r}} &= \frac{\mathbf{r}}{r}, \quad d\mathbf{l}' \times \hat{\mathbf{r}} = \frac{R}{\sqrt{R^2 + y^2}^3} dy \hat{\mathbf{z}} \end{aligned}$$

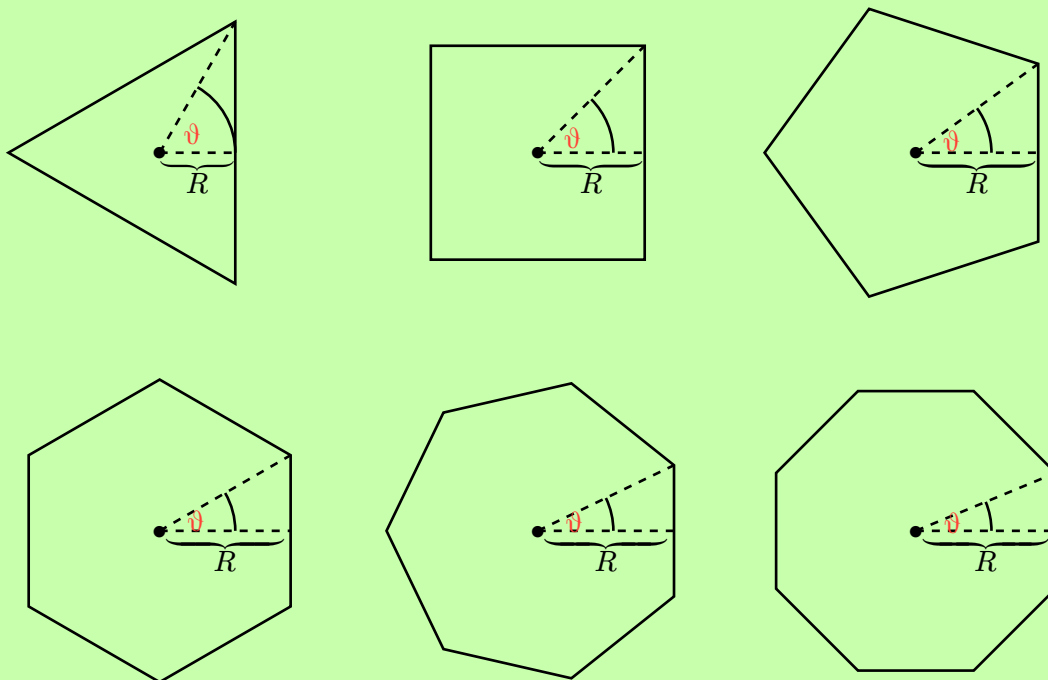
Using these expressions, Biot-Savart law becomes

$$\begin{aligned} \mathbf{B} &= \frac{\mu_0 I}{4\pi} \hat{\mathbf{z}} \int_{-R}^R \frac{R}{\sqrt{R^2 + y^2}^3} dy = \frac{\mu_0 I}{4\pi} \hat{\mathbf{z}} \left[ \frac{Ry}{R^2 \sqrt{R^2 + y^2}} \right]_{-R}^R = \\ &= \frac{2\mu_0 I}{4\pi R \sqrt{2}} \hat{\mathbf{z}} = \frac{\sqrt{2}\mu_0 I}{4\pi R} \hat{\mathbf{z}} \end{aligned}$$

Multiplying this result with 4 (since there are 4 straight wire segments) gives

$$\mathbf{B}(\mathbf{0}) = \frac{\sqrt{2}\mu_0 I}{\pi R} \hat{\mathbf{z}}$$

(b) For a  $n$ -sided polygon, the same principle holds, where if we can find the field from one of the straight segments, the total field is given by multiplying this result with  $n$ .



The total length of the straight right edges for all polygons is

$$L = 2R \tan(\theta), \quad \theta = \frac{\pi}{n}$$

so Biot-Savart becomes

$$\begin{aligned} \frac{R\mu_0 I}{4\pi} \hat{\mathbf{z}} \int_{-R \tan(\frac{\pi}{n})}^{R \tan(\frac{\pi}{n})} \frac{1}{\sqrt{R^2 + y^2}^3} dy &= \frac{R\mu_0 I}{4\pi} \left[ \frac{y}{R^2 \sqrt{R^2 + y^2}} \right]_{-R \tan(\frac{\pi}{n})}^{R \tan(\frac{\pi}{n})} = \\ &= \frac{\mu_0 I \tan(\frac{\pi}{n})}{2\pi R \sqrt{1 + \tan^2(\frac{\pi}{n})}} \hat{\mathbf{z}} \end{aligned}$$

Now,  $\tan(\theta) = \sin(\theta)/\cos(\theta)$  and  $\sqrt{1 + \tan^2(\theta)} = \sec(\theta) = 1/\cos(\theta)$ , so the field from the right line segment is

$$\mathbf{B} = \frac{\mu_0 I \sin(\frac{\pi}{n})}{2\pi R} \hat{\mathbf{z}}$$

Multiplying by the amount of sides for the polygon,  $n$ , gives the total magnetic field

$$\mathbf{B} = \frac{\mu_0 I \sin(\frac{\pi}{n}) n}{2R \pi} \hat{\mathbf{z}} = \frac{\mu_0 I \sin(\frac{\pi}{n})}{2R \frac{\pi}{n}} \hat{\mathbf{z}}$$

(c) As  $n \rightarrow \infty$ , the term  $\frac{\sin(\frac{\pi}{n})}{\frac{\pi}{n}} \rightarrow 1$  (since  $\frac{\pi}{n} \rightarrow 0$  as  $n \rightarrow \infty$ , and we have the standard limit  $\frac{\sin(x)}{x} \rightarrow 1$  as  $x \rightarrow 0$ ). This gives the magnetic field

$$\mathbf{B} = \frac{\mu_0 I}{2R} \hat{\mathbf{z}}$$

which is exactly the field given by a circular loop of radius  $R$ .

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**Problem 5.9**

**Problem 5.10**

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